CSE331 Proofs

Automata and Computability

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1 Midterm

§1.1 Closure Under Union

Theorem (Closure Under Union)
The set of regular languages is closed under union.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.
We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 \cup L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two DFAs

$$D_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$D_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.
We construct a new DFA

$$D = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 \cup L_2$, where:

1. $Q = Q_1 \times Q_2$.
2. Σ is the common input alphabet of D_1 and D_2.
3. The transition function δ is defined as

   $$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

   for all $r_1 \in Q_1$, $r_2 \in Q_2$, and $a \in \Sigma$.
4. $q_0 = (q_1, q_2)$.
5. The set of accepting states is

   $$F = \{(r_1, r_2) \mid r_1 \in F_1 \text{ or } r_2 \in F_2\}.$$

   Equivalently,

   $$F = (F_1 \times Q_2) \cup (Q_1 \times F_2).$$

The DFA D simulates both D_1 and D_2 simultaneously and accepts a string if either D_1 or D_2 accepts it.
Therefore,

$$L(D) = L_1 \cup L_2.$$

Since D is a DFA that recognizes $L_1 \cup L_2$, the language $L_1 \cup L_2$ is regular.
Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under union.}}$$

□

§1.2 Closure Under Complement

Theorem (Closure Under Complement)
The set of regular languages is closed under complement.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.
We need to show that

$$\forall L \in \mathcal{R}_L, \qquad \overline{L} \in \mathcal{R}_L.$$

Since L is regular, there exists a DFA

$$M = (Q, \Sigma, \delta, q_0, F)$$

that recognizes L.
We construct a new DFA

$$M' = (Q', \Sigma', \delta', q_0', F')$$

that recognizes $\overline{L}$, where:

1. $Q' = Q$.
2. $\Sigma' = \Sigma$.
3. $\delta'(q, a) = \delta(q, a)$, for all $q \in Q$ and $a \in \Sigma$.
4. $q_0' = q_0$.
5. $F' = Q - F$.

The DFA M' has the same states, alphabet, transitions, and start state as M, but its accepting and non-accepting states are interchanged.
Therefore,

$$L(M') = \overline{L} = \Sigma^* - L.$$

Since M' is a DFA that recognizes $\overline{L}$, the language $\overline{L}$ is regular.
Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under complement.}}$$

□

§1.3 Closure Under Intersection

Theorem (Closure Under Intersection)
The set of regular languages is closed under intersection.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.
We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 \cap L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two DFAs

$$D_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$D_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.

We construct a new DFA

$$D = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 \cap L_2$, where:

1. $Q = Q_1 \times Q_2$.
2. Σ is the common input alphabet of D_1 and D_2.
3. The transition function δ is defined as

   $$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

   for all $r_1 \in Q_1$, $r_2 \in Q_2$, and $a \in \Sigma$.
4. $q_0 = (q_1, q_2)$.
5. The set of accepting states is

   $$F = \{(r_1, r_2) \mid r_1 \in F_1 \text{ and } r_2 \in F_2\}.$$

   Equivalently,

   $$F = F_1 \times F_2.$$

The DFA D simulates both D_1 and D_2 simultaneously and accepts a string if both D_1 and D_2 accept it.

Therefore,

$$L(D) = L_1 \cap L_2.$$

Since D is a DFA that recognizes $L_1 \cap L_2$, the language $L_1 \cap L_2$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under intersection.}}$$

□

§1.4 Closure Under Difference

Theorem (Closure Under Difference)

The set of regular languages is closed under difference.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 - L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two DFAs

$$D_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$D_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.

We construct a new DFA

$$D = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 - L_2$, where:

1. $Q = Q_1 \times Q_2$.
2. Σ is the common input alphabet of D_1 and D_2.
3. The transition function δ is defined as

   $$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

   for all $r_1 \in Q_1$, $r_2 \in Q_2$, and $a \in \Sigma$.
4. $q_0 = (q_1, q_2)$.
5. The set of accepting states is

   $$F = \{(r_1, r_2) \mid r_1 \in F_1 \text{ and } r_2 \notin F_2\}.$$

   Equivalently,

   $$F = F_1 \times (Q_2 - F_2).$$

The DFA D simulates both D_1 and D_2 simultaneously and accepts a string if D_1 accepts it and D_2 rejects it.

Therefore,

$$L(D) = L_1 - L_2.$$

Since D is a DFA that recognizes $L_1 - L_2$, the language $L_1 - L_2$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under difference.}}$$

□

§1.5 Closure Under Concatenation

Theorem (Closure Under Concatenation)

The set of regular languages is closed under concatenation.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L_1, L_2 \in \mathcal{R}_L, \qquad L_1 \circ L_2 \in \mathcal{R}_L.$$

Since L_1 and L_2 are regular, there exist two NFAs

$$N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and

$$N_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

that recognize L_1 and L_2, respectively.

We construct a new NFA

$$N = (Q, \Sigma, \delta, q_0, F)$$

that recognizes $L_1 \circ L_2$, where:

1. $Q = Q_1 \cup Q_2$, assuming Q_1 and Q_2 are disjoint.
2. Σ is the common input alphabet of N_1 and N_2.
3. The transition function δ is defined as follows:

   For every $q \in Q$ and $a \in \Sigma_\varepsilon$,

$$\delta(q,a) = \begin{cases} \delta_1(q,a), & q \in Q_1,\ q \notin F_1, \\ \delta_1(q,a), & q \in F_1,\ a \neq \varepsilon, \\ \delta_1(q,a) \cup \{q_2\}, & q \in F_1,\ a = \varepsilon, \\ \delta_2(q,a), & q \in Q_2. \end{cases}$$

4. $q_0 = q_1$.
5. $F = F_2$.

The NFA N begins processing the input using N_1. Upon reaching an accepting state of N_1, it may move to the start state of N_2 through an ε-transition and process the remaining input.

Thus, N accepts a string if it can be divided into two parts such that N_1 accepts the first part and N_2 accepts the second part.

Therefore,

$$L(N) = L_1 \circ L_2.$$

Since N is an NFA that recognizes $L_1 \circ L_2$, the language $L_1 \circ L_2$ is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under concatenation.}}$$

□

§1.6 Closure Under Kleene Star

Theorem (Closure Under Kleene Star)

The set of regular languages is closed under Kleene star.

Proof. Let $\mathcal{R}_L$ be the set of all regular languages.

We need to show that

$$\forall L \in \mathcal{R}_L, \qquad L^* \in \mathcal{R}_L.$$

Since L is regular, there exists an NFA

$$N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

that recognizes L.

We construct a new NFA

$$N = (Q, \Sigma, \delta, q_0, F)$$

that recognizes L^*, where:

1. $Q = Q_1 \cup \{q_0\}$, where q_0 is a new state not in Q_1.

2. Σ is the same input alphabet as N_1.

3. The transition function δ is defined as follows:

   For every $q \in Q$ and $a \in \Sigma_\varepsilon$,

$$\delta(q,a) = \begin{cases} \delta_1(q,a), & q \in Q_1,\ a \in \Sigma, \\ \delta_1(q,\varepsilon) \cup \{q_1\}, & q \in F_1,\ a = \varepsilon, \\ \delta_1(q,\varepsilon), & q \in Q_1 - F_1,\ a = \varepsilon, \\ \{q_1\}, & q = q_0,\ a = \varepsilon, \\ \varnothing, & q = q_0,\ a \in \Sigma. \end{cases}$$

4. q_0 is the new start state.

5. The set of accepting states is

$$F = F_1 \cup \{q_0\}.$$

The NFA N introduces a new accepting start state q_0 to accept the empty string ε. It also adds an ε-transition from q_0 to q_1 and from every accepting state of N_1 back to q_1.

Thus, N accepts the empty string or any string formed by concatenating zero or more strings accepted by N_1.

Therefore,

$$L(N) = L^*.$$

Since N is an NFA that recognizes L^*, the language L^* is regular.

Hence,

$$\boxed{\mathcal{R}_L \text{ is closed under Kleene star.}}$$

□

2 Final

Final examination proofs will be added here.